

PROJECT CODE: FKG-BANK

1. PROJECT SUMMARY

This project aims to design a secure, scalable, and manageable network architecture to meet the requirements of a multi-location banking institution. The implementation was carried out in the EVE-NG virtual environment. The network topology consists of Headquarters (HQ), Main Retail Branch, Satellite Branch 1, Satellite Branch 2, and a Remote Worker location. The primary objectives of the project are to establish secure inter-branch communication, achieve network segmentation for resource isolation, enforce access control mechanisms, ensure high availability, and implement fundamental cybersecurity hardening measures to enhance the overall security posture of the network infrastructure.

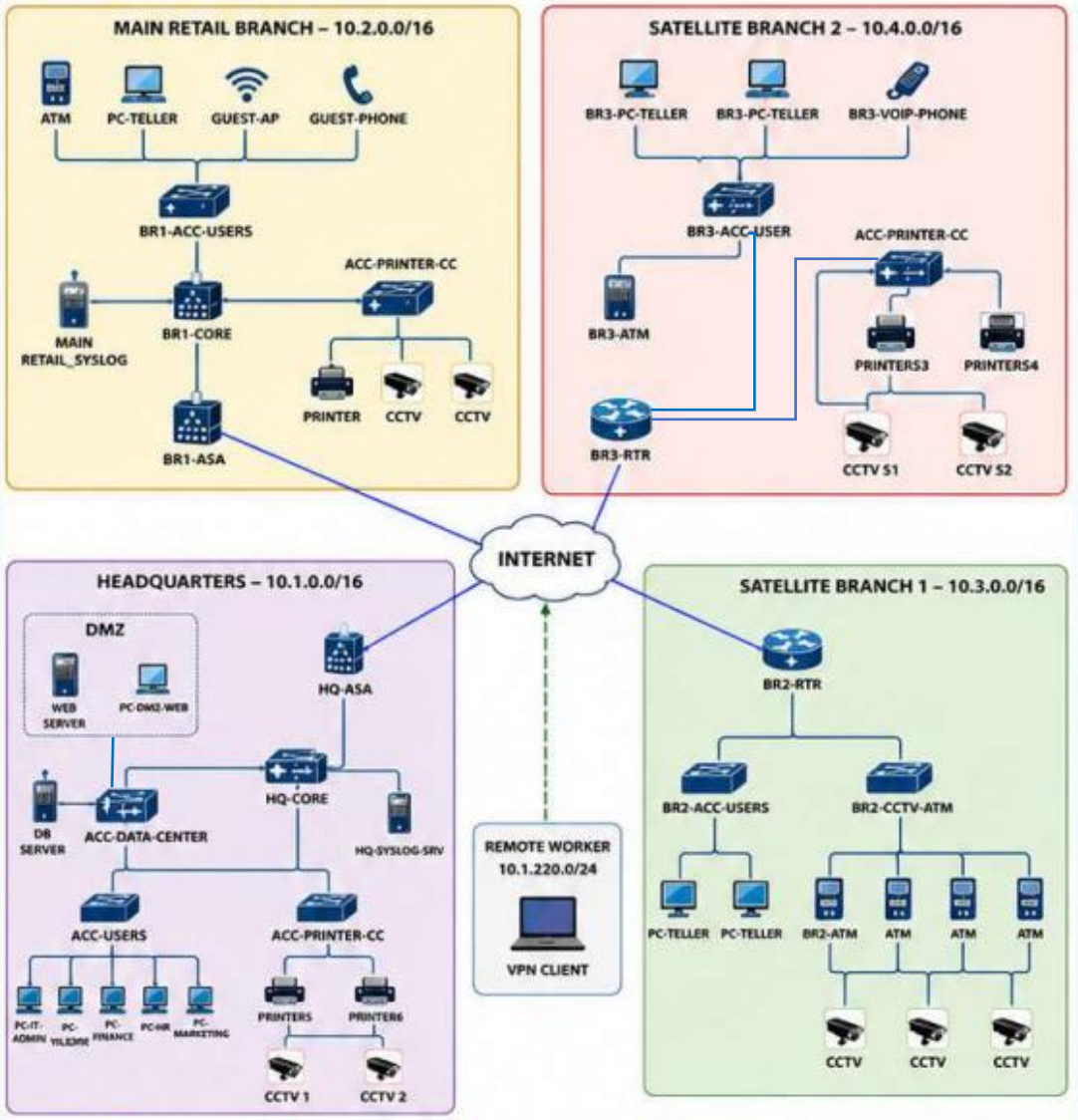
2. OBJECTIVES

- Establish secure communication between branches.
- Implement network segmentation using VLANs.
- Connect headquarters and branch offices through VPN tunnels.
- Restrict unauthorized access using Access Control List (ACL) policies.
- Isolate ATM, CCTV, guest, and user networks to enhance security.
- Validate network connectivity and security controls through comprehensive testing.

3. TECHNOLOGIES

- EVE-NG
- Cisco ASA
- Router ve Layer-2/Layer-3 switching
- OSPF
- IPsec VPN / VTI
- VLAN, ACL, NAT
- Syslog, DMZ, CCTV, Printer, Guest Wi-Fi

4. TOPOLOGY



5. NETWORK HIERARCHY

- Headquarters (HQ): 10.1.0.0/16
- Main Retail Branch: 10.2.0.0/16
- Satellite Branch 1: 10.3.0.0/16
- Satellite Branch 2: 10.4.0.0/16
- Remote Worker Network: 10.1.220.0/24
- All locations are interconnected through the Internet using secure communication channels.

6. SECURITY MECHANISMS

- Site-to-Site VPN tunnels for secure communication
- Dynamic routing using OSPF
- Access control enforcement through ACL policies
- Network Address Translation (NAT) rules
- DMZ hosting Web servers
- Controlled access for ATM systems and others
- Centralized log collection and monitoring using Syslog

7. TEST & VALIDATION RESULTS

Successful establishment and validation of Site-to-Site VPN connectivity.  
Verification of OSPF neighbor adjacencies and route propagation.  
Successful end-to-end connectivity tests between headquarters and branch locations.  
Validation of ACL configurations and access control enforcement mechanisms.  
Confirmation of access restrictions applied to Guest Phone devices and critical network resources.  
Successful testing and verification of DMZ security controls and access policies.

8. CONCLUSION

Within the scope of this project, a secure, scalable, and manageable network architecture for a multi-branch banking environment was successfully designed, implemented, and validated in a simulation environment. By integrating technologies such as VPN, VLAN, ACL, DMZ, and OSPF, secure inter-branch communication and effective protection of network resources were achieved.